

Thermal Infrared Imager for Space Applications

Reproducible Experiment Protocol - based on Krempel, L. (2020), TU Munich master's thesis. This document lists only the experimental setups, materials, and test procedures. Author's results, measured values, discussions and conclusions have been intentionally omitted so that the experiments can be repeated independently and evaluated with fresh data.

1. Overview of Experiments

The thesis describes four main characterization experiments performed on a prototype long-wave infrared (LWIR) micro-bolometer imager intended for a stratospheric balloon flight and later a CubeSat mission. All experiments reuse a common basic setup (Section 2) with variations in parameters described per experiment.

1. Noise Equivalent Temperature Difference (NETD) measurement
2. NETD dependence on detector temperature
3. Temperature dependency of the Signal Transfer Function (SiTF)
4. Absolute temperature accuracy measurement

2. Common Equipment / Materials Needed

2.1 Core Imager Hardware

- Micro-bolometer LWIR detector, e.g. Fraunhofer IMS 17 micrometer pixel-pitch QVGA-IRFPA, sensitive 8-12 micrometer, resolution 256 x 320 px plus blind pixel rows, 16-bit digital raw output, 30 Hz readout
- Detector readout/control board (interfaces detector to a host computer, sets reference voltages)
- Infrared lens/optic thermalized for wide operating temperature range, e.g. f-number 1.0, focal length approx. 8.8 mm, working range 8-12 micrometer
- Focal-plane shutter with two overlapping metal curtains (high-emissivity coating), placed in front of the optics; e.g. product class comparable to a SYIMEXT SU-071 shutter
- Digital temperature sensor mounted on the shutter housing (e.g. Maxim/Analog DS18B20), accuracy approx. 0.5 C
- H-bridge driver circuit plus digital controller to actuate the shutter (reversible polarity, approx. 3.5-4.2 V, 300 mA pulse)
- Thermal control system: one or two Peltier elements thermally coupled via copper plates/heat sink to the detector board
- Precision resistance temperature sensor (e.g. PT1000) read out via a Wheatstone bridge, reference resistors, a low-noise voltage reference and a 24-bit ADC (e.g. AD7193), plus a low-resolution reference sensor for one-point calibration
- Programmable power supply to drive the Peltier element under digital control
- Central control and data-handling unit (e.g. Raspberry Pi) with attached data storage, running the readout, shutter control and PID thermal-control software
- Calibrated black-body radiation source with known emissivity (e.g. approx. 0.95) and stable, adjustable surface temperature (commercial unit such as a Voltcraft IR-350 class device)
- General lab equipment: optical bench/mount for aligning detector, optics, shutter and black body along one

optical axis; cabling and power supply

2.2 Things to Watch Out for (General)

- Detector temperature must be fully stabilized before any measurement or before adjusting the reference voltage V_{ref1}
- Close the shutter before changing the thermal-control set point or before re-adjusting V_{ref1}
- An offset (shutter) calibration is only valid for a short time (on the order of 30 seconds); repeat it if the detector temperature changes, the reference voltage is adjusted, or ambient stray-radiation conditions change
- The black body needs a long settling time after a set-point change (up to approximately 30 minutes) before its surface is stable
- When the black body is placed very close to the optics (shorter than the lens minimum working distance), the image will be blurred; make sure the target area extends beyond the image field by at least the blur radius
- Keep the imaged scene static during multi-frame averaging, since motion between frames reintroduces noise/artifacts
- Dust or contamination on the detector window will locally reduce responsivity and should be checked/cleaned before precision tests
- Record ambient/room temperature, as it affects the achievable thermal-control range of the Peltier stage

3. Basic Experimental Setup (Used by All Experiments)

Setup Steps

1. Power the detector readout board and connect it to the control/data-handling unit.
2. Manually align the optics in front of the detector.
3. Install the shutter directly in front of the optics.
4. Place the calibrated black-body radiation source in front of the assembled imager, aligned on the optical axis.
5. Turn on the thermal control and set the desired detector temperature for the experiment.
6. Wait until the detector temperature has stabilized.
7. Perform the detector startup procedure (see Section 4) to set the reference voltage V_{ref1} .
8. Adjust the position of the black body and optics so that all detector pixels view the controlled-temperature surface, accounting for blur if the source is closer than the lens minimum working distance.
9. Turn on the black-body source and close the imager shutter before beginning measurements.

Procedure to Change Detector Temperature Mid-Experiment

1. Close the shutter.
2. Set the thermal control system to the new target temperature.
3. Monitor the detector temperature until it has stabilized.
4. Re-set the reference voltage V_{ref1} following the startup procedure appropriate for the experiment.

4. Detector Startup / Reference Voltage Procedure

This procedure must be completed before taking data in any experiment, and repeated whenever the detector temperature changes.

1. Ensure the detector temperature is already stabilized.
2. Close the shutter to create a homogeneous radiation scene.

3. Choose one of two methods to set the reference voltage V_{ref1} : (a) an iterative half-scale adjustment, or (b) applying a predefined voltage value known for that detector temperature.
4. If using the iterative method: set V_{ref1} , capture detector output, compute the average raw value, and adjust V_{ref1} until the average is close to half of the full 16-bit scale.
5. If using a predefined value: set the known V_{ref1} for the target temperature, then verify the resulting average output does not deviate by more than about 10 percent from half-scale.

Watch out: which method to use depends on the specific experiment being run; use the same method consistently within a data set for comparability.

5. Experiment 1: Noise Equivalent Temperature Difference (NETD)

What Is Tested

The noise performance of the imager is characterized by measuring the NETD, i.e. how small a temperature difference the detector can resolve above its own noise floor.

Materials Needed

- Full basic setup (Section 3)
- Black-body radiation source set to a fixed known temperature
- Shutter used as a second, homogeneous reference temperature scene (instead of a second black-body setting, since black-body set-point changes take too long)

Procedure

1. Set the detector temperature (e.g. to a chosen fixed value) and perform the startup/ V_{ref1} procedure via half-scale adjustment.
2. Set the black-body source to a fixed target temperature.
3. Open the shutter and capture 50 consecutive images of the black-body scene.
4. Close the shutter and note its temperature (read from the shutter's built-in temperature sensor).
5. Capture another 50 consecutive images with the shutter closed.
6. Do not change the reference voltage V_{ref1} between the two data sets.

Watch Out For

- The black body cannot be used for both temperature scenes in this test because changing its set point takes roughly 30 minutes, during which detector drift makes the two data sets incomparable
- Note the exact shutter temperature for each run since it is used as the reference low-temperature scene
- Keep all other detector settings identical across both 50-image captures